Maternal Tobacco Smoke and Fetal Growth

Active maternal tobacco smoking is associated with lower birthweight and higher preterm risk, with dose-response evidence; this is a birth-size/preterm claim, not a personality or IQ card.

GRADE: MODERATE | STATUS: DRAFT | DOMAIN: PREGNANCY | REPLICATION: VERIFIED

WHAT WE KNOW (Narrowest Supported Conclusion)

Active maternal cigarette smoking is associated with reduced fetal growth and higher preterm risk, with a reported dose-response. This is a birth-size and gestational-duration claim. It is not a personality or IQ card.

EVIDENCE AND CAUSAL ASSESSMENT

Kramer (1987) assessed 43 determinants across 895 studies: in developed-country settings, cigarette smoking was the most important identified modifiable determinant of low birthweight, with effects on both growth restriction and gestational duration. Later biomarker work (not catalogued here as a second ID) often shows tighter dose-response with cotinine than with self-report. This remains epidemiology plus dose-response, not an RCT of smoking assignment.

EFFECT SIZE AND PERSISTENCE

Birthweight deficits are typically on the order of 150–250 g for active smoking in older syntheses, larger at higher consumption. Persistence of size difference into later life is mixed and confounded; do not carry the birthweight effect into IQ.

BIAS, HETEROGENEITY, AND CONFOUNDING

GI attenuation: partially confounded (20–50%) (maternal smoking during pregnancy shows partial attenuation in sibling and Mendelian randomization studies).

SES, co-exposures, partner smoking, nutrition, residual confounding. Genetically informed designs for this specific path were not in the Chapter 2 dossier.

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Does not mean a smaller baby has a lesser future, or that smoking is the only growth constraint.

WHAT ACTION IS REASONABLY JUSTIFIED

Population and clinical support for smoking cessation in pregnancy is justified on birthweight/preterm grounds. Treating birth size as destiny is not.

VERIFIED SOURCES & DOI IDENTIFIERS

- Kramer (1987), Bulletin of the World Health Organization